Project Planning & Management

End-to-End Business Analytics Decision Platform – Tourism & Hospitality Initiative

DEPI – Digital Egypt Pioneers Initiative | Track: Data Science

Trainer: Sara Ashraf

Repository: https://github.com/marioma-123/business-analytics-decision-platform

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1. Project Proposal

1.1 Overview

This project builds an end-to-end analytics and decision platform for a tourism and hospitality business that manages short-term rental accommodation across several locations. It integrates data from Inside Airbnb, World Bank World Development Indicators, Wikipedia, and the Open-Meteo historical weather API; analyses drivers of prices, demand indicators, and guest satisfaction; predicts nightly listing prices with machine learning; and delivers results through an interactive dashboard and a deployed prediction service.

1.2 Problem Statement

A company has data scattered across multiple sources but lacks an integrated analytical solution for monitoring performance, understanding its drivers, and predicting future outcomes. Short-term rental listing records, historical weather, country-level economic indicators, and web-collected landmark/location information are stored separately. Management therefore cannot easily:

- Monitor prices, availability, and ratings across locations and over time.
- Understand drivers of price and demand indicators, including location, property/room type, season, weather, and guest reviews.
- Predict nightly price per listing to support pricing and planning decisions.

1.3 Project Objective

Build a complete data science solution following the CRISP-DM lifecycle, from business understanding through deployment and communication.

- Define the business problem, stakeholders, KPIs, analytical questions, and success criteria.
- Collect data from a SQL database (Inside Airbnb files), REST APIs (Open-Meteo and World Bank indicators), and Wikipedia web scraping.
- Integrate and clean data into an analysis-ready dataset.
- Explore descriptive statistics, distributions, correlations, and grouped analyses.
- Build an interactive dashboard with executive KPIs, trends, comparisons, and geographic visualization.
- Train and compare Linear/Ridge Regression, Random Forest, and Gradient Boosting models to predict nightly listing prices.
- Evaluate models and justify the final selection and limitations.
- Track experiments with MLflow, serve predictions with FastAPI, package with Docker, automate checks with GitHub Actions, and deploy to a cloud platform (provider undecided).
- Communicate findings through an executive summary, technical report, dashboard, and stakeholder presentation.

1.4 Scope

CRISP-DM phase	Deliverable
1. Business Understanding	Business problem, stakeholders, KPIs, analytical questions, success criteria
2. Data Collection	SQL database (Inside Airbnb), REST APIs (Open-Meteo and World Bank), Wikipedia scraping
3. Data Preparation	Integration, missing values, formatting, normalization, categorical transformation
4. EDA	Descriptive statistics, distributions, correlations, group-by analysis
5. Visualization	Executive KPIs, trend and comparative charts, map, interactive dashboard
6. Machine Learning	Multiple regression models for nightly price prediction
7. Evaluation	Metrics, model comparison, limitations, selection rationale
8. Deployment	MLflow, FastAPI, Docker, GitHub Actions, cloud hosting (provider undecided)
9. Communication	Executive summary, technical report, dashboard, stakeholder presentation

1.5 Stakeholder

Stakeholder	Role	Main interest / need
General Manager / Executive	Decision maker	Overall performance KPIs and trends across locations
Revenue / Pricing Manager	Primary user of model	Price drivers and price predictions to set competitive rates
Marketing / Destination Manager	Planner	Which locations, segments, and seasons perform best
Property Managers	Daily user	Performance and guest ratings by property and location
Hosts / Property owners	Indirect user	Guidance on realistic pricing
Guests / Travelers	Indirect beneficiary	Better matched, fairly priced offers
Data / IT team	Maintainer	Reproducible pipeline, deployed service, documentation
DEPI Trainer / Evaluation Committee	Evaluator	Deliverables, quality, individual contribution

1.6 Data Sources

#	Source	Type	What it provides	Access / license
1	Inside Airbnb – https://insideairbnb.com/get-the-data/	CSV into SQL	Listings, calendar availability, reviews for Barcelona, Amsterdam, Cape Town, Athens	Public download; follow site license terms
2	World Bank WDI – https://databank.worldbank.org/source/world-development-indicators	REST API / file	GDP per capita, tourism arrivals, consumer price inflation	Public; World Bank terms
3	Wikipedia	Web scraping	Landmarks and coordinates; distance-to-nearest-landmark features	Public; follow terms and credit source
4	Open-Meteo Historical Weather API – https://archive-api.open-meteo.com	REST API	Daily historical weather by latitude/longitude and date	Public API; follow terms

Selected cities: Barcelona (Spain), Amsterdam (Netherlands), Cape Town (South Africa), and Athens (Greece). Backup cities: Rome (Italy) and Lisbon (Portugal).

1.7 Methodology

CRISP-DM: Business Understanding → Data Understanding/Collection → Data Preparation → Modeling → Evaluation → Deployment/Communication. Work is tracked in GitHub Projects. Code is managed with feature branches and pull requests.

1.8 Expected Deliverables

A professional portfolio repository containing: Business Case, Data Sources, SQL, Data Collection, Data Cleaning, EDA, Visualization, Machine Learning, Model Evaluation, Dashboard, Deployment, Documentation, and Stakeholder Presentation.

1.9 Assumptions & Constraints

- The team has six members.
- Only publicly available data sources are used.
- Inside Airbnb prices are asking prices, not confirmed transaction prices. Demand is represented by indirect indicators (availability and review counts), not actual bookings.
- World Bank indicators are country-level; with four countries, each indicator has few distinct values and mainly provides economic context and a coarse model feature.
- Deployment will use free tiers or student credits where possible. Cloud provider/account access is undecided.
- Final project discussion date is unknown and will be added when the training schedule is confirmed.

2. Project Plan

2.1 Milestones

Dates follow the DEPI schedule in the Documentation Guidelines.

Milestone	Description	Due date
M1	Planning & Management, Literature Review, Requirements Gathering submitted on GitHub	16 Oct 2026
M2	System Analysis & Design submitted	6 Nov 2026
M3	Implementation (source code & execution, including deployment) complete	30 Nov 2026
M4	Final Presentation, Testing & Reports submitted	4 Dec 2026
M5	Final project discussion	Unknown – pending training schedule

2.2 Project Timeline

- Documentation and technical work intentionally overlap.
- Setup: choose cities and confirm data sources; set up repository and roles; notify trainer.
- Documentation: planning, literature review, requirements; system analysis and design; testing/QA documentation; user manual, presentation, infographic.
- Technical work: Phase 2 Data Collection; Phase 3 Data Preparation; Phase 4 EDA; Phase 5 Visualization & Dashboard; Phase 6 Machine Learning with MLflow; Phase 7 Evaluation; Phase 8 Deployment (API, Docker, CI, cloud).
- Timeline markers in the original plan: 11 Oct, 18 Oct, 25 Oct, 1 Nov, 8 Nov, 15 Nov, 22 Nov, and 29 Nov; M1–M4 dates are listed above.

2.3 Deliverables per Phase

Phase	Deliverable	Owner	Due
Phase 1	Business case (problem, KPIs, questions, success criteria)	Mariam – Team Lead	16 Oct 2026
Phase 2	SQL scripts + collection notebooks + data sources documentation	Asmaa – Data Engineer	24 Oct 2026
Phase 3	Cleaning notebook + processed dataset	Rofida – Data Preparation, QA CI	&3 Nov 2026
Phase 4	EDA notebook + findings summary	Rofida,Asmaa – Research Analyst	10 Nov 2026
Phase 5	Dashboard	Mariam – Visualization & UX	20 Nov 2026
Phases 6–7	ML notebook + MLflow runs +Feature Engineering	Ahmed,Ahmed,Mostafa – ML Engineer	27 Nov 2026
Phase 8	FastAPI app, Dockerfile, GitHub Actions workflow, cloud link, evaluation report	Ahmed , Ahmed, Mariam,Mostafa	30 Nov 2026
Phase 9	Executive summary, technical report, presentation	Mariam (coordination); all members contribute	4 Dec 2026

2.4 Resource Allocation

Resource	Details / status
People	6 team members
Languages / libraries	Python, SQL, pandas, scikit-learn
Database	PostgreSQL
Dashboard tool	Power BI
MLOps / Deployment	MLflow, FastAPI, Docker, GitHub Actions; cloud provider and account/credits undecided
Collaboration	GitHub, WhatsApp, GitHub Projects
Data	Inside Airbnb, World Bank WDI, Wikipedia, Open-Meteo
Compute	Local machines

2.5 Team Workflow

- Weekly meeting: Friday at 9:00 PM, 30–45 minutes, to review progress, blockers, and next week's tasks.
- Pull requests: each PR is reviewed by at least one other member before merging.
- Commit messages: short and meaningful (e.g., “add: median imputation for price”).
- Trainer reporting: weekly deliverables are submitted as agreed with the trainer.

3. Task Assignment & Roles

Every member has a technical responsibility because each member's contribution is presented and evaluated individually. Backup assignments have not been provided and are marked as not assigned.

#	Member	Role	Technical responsibility	Backup
1	Mariam	Team Lead & Business Analyst	Phase 1;Business case, Phase 5	Not assigned
2	Mostafa	ML Engineer	Phase 6,7,8	Not assigned
3	Ahmed Hussein	ML Engineer	Phase 6,7,8	Not assigned
4	Ahmed	ML Engineer	Phase 6,7,8	Not assigned
5	Rofida	Data Engineer	Phase 3 and 4	Not assigned
6	Asmaa	Data Engineer (SQL & Collection)	Phase 2,4	Not assigned

Shared responsibility: architecture, deployment diagram, DFD, and sequence/activity diagrams (Mariam, Asmaa, and Ahmed). Each member prepares and presents their own part in the final presentation.

4. Risk Assessment & Mitigation Plan

Probability and impact are rated Low / Medium / High.

ID	Risk	Prob.	Impact	Mitigation	Owner
R1	Late start / lost time before first deadline	High	High	Parallel documentation and technical work; daily check-ins until 16 Oct	Mariam
R2	Data source unavailable, rate-limited, or changes format	Medium	High	Save raw snapshots; identify a backup source	Asmaa
R3	Wikipedia scraping blocked, or landmark pages differ by city	Medium	Medium	Scrape politely with delays; check terms; use the official Wikipedia API or a manually verified landmark list per city	Asmaa
R4	Data quality issues (missing values, inconsistent formats, outlier prices)	High	Medium	Document cleaning decisions; compare imputation/outlier options	Ahmed Nassar
R5	Sources cannot be joined due to missing common keys	Medium	High	Verify join keys (city, date, latitude/longitude) before building the pipeline	Asmaa
R6	Models perform poorly	Medium	Medium	Try several models and feature sets; report honestly with limitations	Ahmed
R7	Member unavailable or not contributing	Medium	High	Assign backups; check weekly progress; inform trainer early	Mariam
R8	Merge conflicts / lost work	Medium	Medium	Feature branches, small PRs, frequent commits	Mariam
R9	Secrets (API keys, cloud credentials) pushed to GitHub	Medium	High	Use .gitignore, environment variables, and GitHub Secrets	Everyone
R 10	Deployment consumes time needed for core work	High	High	Start Phase 8 only after a working model exists; keep it minimal (one prediction endpoint, one container)	Mariam
R 11	Cloud account, credits, or free-tier limits block deployment	Medium	High	Confirm access early; keep Docker-based local deployment as fallback; document both	Mariam
R 12	Deployed service down or slow on presentation day	Medium	High	Test one day before; keep a recorded demo and local run as backup	Ahmed
R 13	World Bank country-level indicators add little signal	High	Medium	Use year-varying values where possible and report their contribution honestly	Asmaa
R 14	Listed prices differ from real transaction prices; demand not directly observable	High	Medium	Document as a limitation; use availability and review counts only as indirect indicators	Mostafa
R 15	Large Inside Airbnb files slow processing or exceed GitHub limits	Medium	Medium	Limit cities; keep large raw files out of the repository (.gitignore); document download steps	Asmaa

5. Project KPIs

These measure project success. Business KPIs (e.g., average price and rating trends) belong in the Business Case. Targets are proposed by the team and may be revised.

Category	KPI	How to measure	Target
Delivery	On-time submission of milestones	Milestones submitted by due date / total milestones	100%
Delivery	Task completion	Completed tasks / planned tasks (weekly)	≥ 90%
Collaboration	Contribution balance	Every member has meaningful commits and PRs	All 6 members
Data	Source integration	Sources successfully integrated	4 of 4 sources
Data	Data completeness after cleaning	% non-missing in key columns	≥ 95%
Modeling	Models compared	Number of models trained and evaluated	≥ 3
Modeling	Model performance	R^2, RMSE, MAE vs. a simple baseline	Better than baseline
Modeling	Experiment tracking	Share of model runs logged in MLflow	100%
Dashboard	Required views present	KPIs, trends, comparisons, map, filters	All present
Dashboard	Load time	Seconds to load main view	≤ 5 s
Deployment	Prediction API response time	Average seconds per request	≤ 1 s
Deployment	Service availability	Reachable during demo/evaluation period	Reachable
Deployment	CI pipeline health	GitHub Actions checks pass on merges to main	Passing
Reproducibility	Fresh-install run	A teammate runs the project from README (including Docker)	Succeeds
Documentation	Coverage	Required documents completed / required documents	100%

Open Items / To Confirm

- Cloud hosting provider and whether a student/free-tier account or credits are available.
- Final project discussion date, once the DEPI training schedule is confirmed.
- Backup person for each team member, if backups are to be assigned.

Literature Review: Comparable Airbnb Pricing Projects

The following table reproduces the information in the supplied comparison image as a real, selectable text table. Question marks and missing-feature marks are retained where the supplied comparison did not confirm a capability.

Dimension	TravelPulse (planned)	amayumradia	HF Copenhagen
Type	End-to-end platform	Academic project (Harvard Project)	Small API app
City / scope	4 cities in 4 countries	New York	Copenhagen
Data sources	4 (Inside Airbnb + weather + World Bank + Wikipedia)	Inside Airbnb	?
Models	Ridge, Random Forest, Gradient Boosting	Linear, Ridge, Lasso with k-fold	? (has a “Make prediction” step)
Distinctive feature	Weather + landmarks + economic indicators	Seasonality premium	Calculates a price range around the predicted price
EDA	✓	✓ (about 3 notebooks)	?
Dashboard	✓ Power BI	✗	✗
API	✓ FastAPI	✗	✓ FastAPI
Docker	✓	✗	✓ (Python 3.9-slim, port 7575)
MLflow	✓	✗	?
CI/CD	✓ GitHub Actions	✗	?
Documentation	Business case + report + presentation + manual	Short README + literature review + docx report	?
Project management	Risks, KPIs, roles, timeline	✗	✗
Tech stack	Modern Python	README requires Python 2	Python 3.9

Note: “TravelPulse (planned)” describes the intended scope in the project plan, not verified completed functionality. A question mark means the supplied comparison did not establish the answer; it is not a claim that the feature is absent.

Project links

amayumradia — AirBnB pricing prediction: https://github.com/amayumradia/AirBnB-pricing-prediction

HF Copenhagen — FastAPI Airbnb app: https://huggingface.co/insightful-stays/airbnb-improvement-generator

Analysis and Lessons for TravelPulse

1. amayumradia — Airbnb pricing prediction

Based on the supplied comparison, this academic project focuses on Airbnb pricing in New York and uses Linear, Ridge, and Lasso regression with k-fold validation. Its distinctive element is a seasonality premium. The supplied table does not list a dashboard, API, Docker, MLflow, or CI/CD. Treat these as unlisted in the comparison, not as independently verified absences.

2. HF Copenhagen — prediction API

The supplied comparison characterizes this project as a small API application using FastAPI, Docker, and Hugging Face Spaces deployment. It also notes a prediction step that calculates a price range around the predicted price. The comparison leaves its data sources, model details, EDA, MLflow, and CI/CD uncertain; those points should be checked against the linked project before final submission.

3. Planned contribution of TravelPulse

TravelPulse is planned as a broader analytics and decision platform: four cities across four countries; Inside Airbnb, historical weather, World Bank indicators, and landmark/location information; EDA and a Power BI dashboard; multiple regression/tree-based models; MLflow experiment tracking; FastAPI, Docker, and GitHub Actions. This combination is the intended distinction from a single-city pricing notebook or a small prediction API. These are planned capabilities and should not be presented as completed results until implemented and tested.

4. Lessons to apply

- Evaluate whether seasonal and weather features improve validation performance rather than assuming they do.
- Compare candidate models with a simple baseline and report MAE, RMSE, and R^2.
- Make the API reproducible and document expected inputs, outputs, and error cases.
- Keep the distinction between asking prices and completed transaction prices explicit; availability and review counts are indirect demand indicators, not booking records.
- State the limits of country-level economic indicators and avoid overstating their predictive value.

References

[1] amayumradia. *AirBnB pricing prediction*. GitHub repository. https://github.com/amayumradia/AirBnB-pricing-prediction

[2] RJuro. *fastapi-airbnb-bds-class*. Hugging Face Spaces, commit cda936826393524001cbeb3e6841fdedf2bae8f9. Open referenced commit

Review note: this comparison is based on the team-provided table. Verify project features from the linked sources and add any further sources consulted before submitting the final literature review.